

NEET Chemistry

Detailed NEET-UG Syllabus

Exam Coverage	NEET-UG aligned topic map for complete subject planning
Format	Unit-wise syllabus with detailed bullet-point subtopic breakdown
Use Case	Course brochure, batch syllabus design, chapter planning and academic structuring

Structured Syllabus

Unit 1: Some Basic Concepts of Chemistry

- Mole concept, stoichiometry, atomic and molecular masses, empirical and molecular formula.
- Limiting reagent, concentration terms, percentage composition and chemical calculations.

Unit 2: Atomic Structure

- Bohr's model, dual nature of matter, quantum mechanical model and quantum numbers.
- Electronic configuration and stability of half-filled and completely filled orbitals.

Unit 3: Classification of Elements and Periodicity

- Modern periodic table and periodic trends in atomic radius, ionization enthalpy, electron affinity and electronegativity.
- Valence and periodicity in physical and chemical properties.

Unit 4: Chemical Bonding and Molecular Structure

- Ionic, covalent and coordinate bonding, Lewis structures, VSEPR theory and hybridization.
- Resonance, bond parameters, hydrogen bonding and molecular orbital concepts.

Unit 5: States of Matter

- Gaseous and liquid states, gas laws, kinetic theory, ideal and real gases.
- Intermolecular forces, liquefaction of gases and critical temperature.

Unit 6: Thermodynamics

- System and surroundings, internal energy, enthalpy, entropy and Gibbs free energy.
- First law of thermodynamics, Hess's law and spontaneity of reactions.

Unit 7: Equilibrium

- Chemical equilibrium, law of mass action, equilibrium constant and Le Chatelier's principle.
- Ionic equilibrium, pH, buffer solutions, hydrolysis and solubility product.

Unit 8: Redox Reactions

- Oxidation number, balancing redox equations and redox titration principles.

Unit 9: Hydrogen

- Position of hydrogen, isotopes, hydrides, water, hydrogen peroxide and their properties.

Unit 10: s-Block Elements

- General properties and important compounds of alkali and alkaline earth metals.

Unit 11: p-Block Elements

- General introduction and detailed chemistry of important groups including boron, carbon, nitrogen, oxygen and halogens.
- Preparation, properties and uses of major compounds of p-block elements.

Unit 12: Organic Chemistry – Basic Principles

- Classification and nomenclature, isomerism, electronic displacement effects, reaction intermediates and mechanism basics.

Unit 13: Hydrocarbons

- Alkanes, alkenes, alkynes and aromatic hydrocarbons — preparation, properties and reactions.

Unit 14: Environmental Chemistry

- Air, water and soil pollution, atmospheric chemistry and strategies for environmental control.

Unit 15: Solid State

- Types of solids, crystal lattice, unit cell, packing efficiency, defects and electrical properties.

Unit 16: Solutions

- Concentration terms, Raoult's law, ideal and non-ideal solutions, colligative properties and abnormal molecular mass.

Unit 17: Electrochemistry

- Conductance, electrochemical cells, Nernst equation, batteries, corrosion and electrolysis.

Unit 18: Chemical Kinetics

- Rate of reaction, factors affecting rate, order and molecularity, integrated rate equations and Arrhenius equation.

Unit 19: Surface Chemistry

- Adsorption, catalysis, colloids and emulsions with practical examples.

Unit 20: General Principles and Processes of Isolation of Elements

- Occurrence of metals, concentration of ores, extraction, refining and thermodynamic principles in metallurgy.

Unit 21: d- and f-Block Elements

- Transition elements, lanthanoids and actinoids, oxidation states, coloured ions and magnetic properties.

Unit 22: Coordination Compounds

- Nomenclature, isomerism, bonding theories, stability and applications of coordination compounds.

Unit 23: Haloalkanes and Haloarenes

- Preparation, physical and chemical properties, substitution and elimination reactions.

Unit 24: Alcohols, Phenols and Ethers

- Preparation, reactions, acidity/basicity and distinguishing features of these functional groups.

Unit 25: Aldehydes, Ketones and Carboxylic Acids

- Preparation, nucleophilic addition reactions, oxidation-reduction behaviour and acidity trends.

Unit 26: Organic Compounds Containing Nitrogen

- Amines, diazonium salts, preparation, properties and important reactions.

Unit 27: Biomolecules

- Carbohydrates, proteins, enzymes, vitamins and nucleic acids.

Unit 28: Polymers

- Classification, polymerization reactions and important commercial polymers.

Unit 29: Chemistry in Everyday Life

- Drugs, detergents, food preservatives, sweeteners and common chemical applications in daily life.